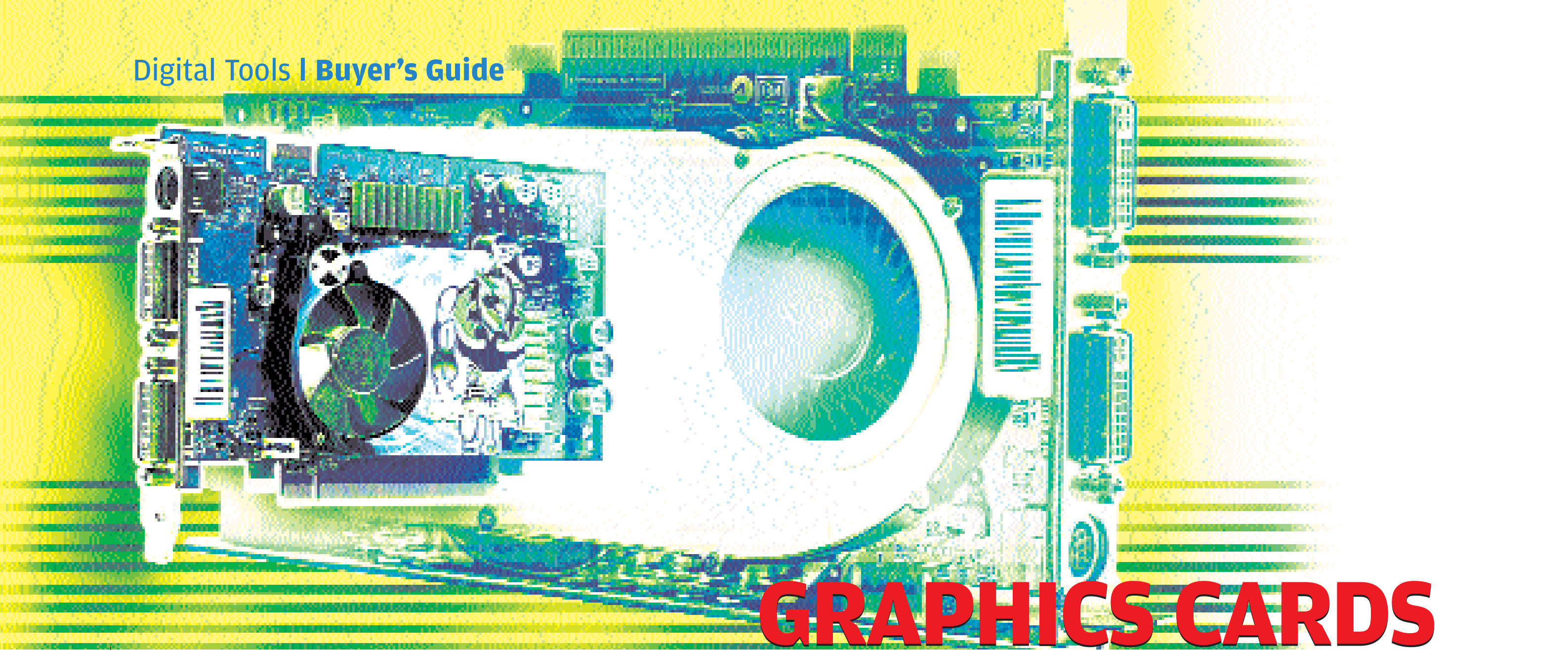

GRAPHICS CARDS

The graphics card (aka display card or video adapter) plays a very important role in bridging the communication gap between a computer and the user by helping display the computer's output information on a monitor. At least that was what graphics cards were meant to do when they were first invented. Today, though this is the primary function of a graphics card, there are many other things that depend on it, such as 3D capabilities, multiple displays, TV-out, etc. A graphics card today can be one of the most expensive components of a PC.

Questions To Ask The Dealer

How many years is the warranty period?
While most vendors offer a three-year warranty, some don't. Check for this.

What interface does the card have?
There are two types of interfaces available, AGP and PCIe. You should make sure what type of graphics card you need.

What accessories come with it?
Some graphics cards have a video capture facility, whereas some are bundled with a TV-Tuner. Such extra features require extra accessories such as cables, a remote control, etc., and you should check if the necessary accessories are bundled. Most graphics cards also bundle a free game.

Is there proper driver support?
Proper driver support will keep the graphics card in tune with the latest applications and operating systems. Certain vendors provide special features in their drivers such as support for 3D goggles, which are not available in generic drivers.

Usage Tips

- Check for updated drivers periodically on the manufacturer's Web site. These address many bugs, and sometimes provide such a performance boost that you may liken it to a free upgrade.
- Keep the graphics settings to the defaults to improve system performance.
- Change the display settings in-game where possible.
- Keep the space near the air intake of the graphics card clear of obstacles to facilitate efficient cooling.
- There are many utilities available to overclock graphics chips and memories. This will give a small performance boost in 3D games.

The Future

It is an undisputable fact that today, the gaming industry is the driving force behind the graphics card industry. Graphics cards have evolved into much more than display adapters. Already, there are systems with multiple graphics cards—the technologies for this are SLI from nVidia and CrossFire from ATi. These claim to almost double 3D graphics performance. Then there are some who put two graphics chips on the same graphics card to achieve similar results.

Looking at the many similarities with the CPU, GPUs, as graphics chips are known today, will move on to newer fabrication processes, and more transistors will be packed onto the chip. The chip will therefore consume less power, and will run at cooler temperatures. Newer cards will ship with the amazingly fast GDDR6, which will provide a humongous bandwidth increase.

Microsoft will soon be moving on to DirectX 10, and Shader Model 4 will be the next frontier that graphics chip makers will keep in mind when designing their next-generation chips. ■

Points To Ponder

What type of graphics card will my system support?
Consult your motherboard manual to find out if you need an AGP or a PCIe card.

What type of applications do I run?
If you primarily run 2D applications, any graphics card will be good enough, but for 3D applications and gaming, a 3D graphics card will be required.

How powerful a graphics card will my computer support?
A high-end graphics card will work with any compatible system, but in order to get the most juice out of it, a powerful CPU and a good amount of memory has to complement it. Similarly, a low-end card will not give great performance even when used with a high-end system. Also, some high-end graphics cards consume a great deal of power, and the system's power supply must be powerful enough to cope with its power needs.

What should be the compliance requirement of the 3D graphics card?
Newer games require an OpenGL 2.0 or DirectX 9.0c compatible graphics card to run. Buy a graphics card that supports the highest version of OpenGL and DirectX.

What type of outputs should the graphics card have?
If you have a digital display, the graphics card should ideally have a DVI-Out. Similarly, if you are keen on big-screen gaming, you'll need a TV-out.